UNIVERSITY OF RUHUNA
DEPARTMENT OF ELECTRICAL AND INFORMATION ENGINEERING
EE 3306 – Signals and Systems
Laboratory 3

Analog / Digital conversion

INTRODUCTION: This laboratory will demonstrate the basic operation of an A/D converter and D/A converter.

EDUCATIONAL OBJECTIVES:

1. Understand the operation of an Analog to Digital Converter
2. Understand the operation of a Digital to Analog Converter

PRE-LAB QUESTIONS:

1. Explain about the analog to digital conversion.
2. What is the function of PWM in D/A conversion?
3. What are the Applications of Digital to Analog Converters?

COMPONENT NEEDED:

- Oscilloscope
- Dual Power Supply
- OP AMP 741
- Encoding Unit
- Resistors: 1kΩ, 1.5kΩ, 2kΩ, 4kΩ
- Bread Board

BACKGROUND AND THEORY

There are many system problems that require connecting a digital portion of the system to an analog component. This 'meeting of circuits' is called an interface. There are two problems associated with this interface that must be met.

a) The conversion of digital signal to analog.
b) The conversion of analog signal to digital.

Any analog voltage can be expressed as a binary word by assigning voltage weights to each bit positions. If we consider a four bit word, voltage values of 8, 4, 2and 1 could be assigned to each bit positions as shown Table 1.

Table 1: Expressed as a binary word by assigning voltage

Binary Word	Voltage
0000	0
0001	1
0010	2
0011	3
0100	4
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1101	13
1110	14
1111	15

Each binary count represents 1/15th of the entire voltage. Resolution of an n bit system is given by,

$$\text{Res\%(Q)} = \frac{V_{ref}}{2^N - 1} \text{ X } 100\%$$

N- Number of bits

Sample, hold unit and a quantizer together form the analog to digital converter. Sample and hold device takes a snapshot of the analogue signal periodically and then holds that value constant for the period. The quantizer calculates the binary number.

An input output characteristics for a 3-bit quantizer is illustrated in the Figure 1.

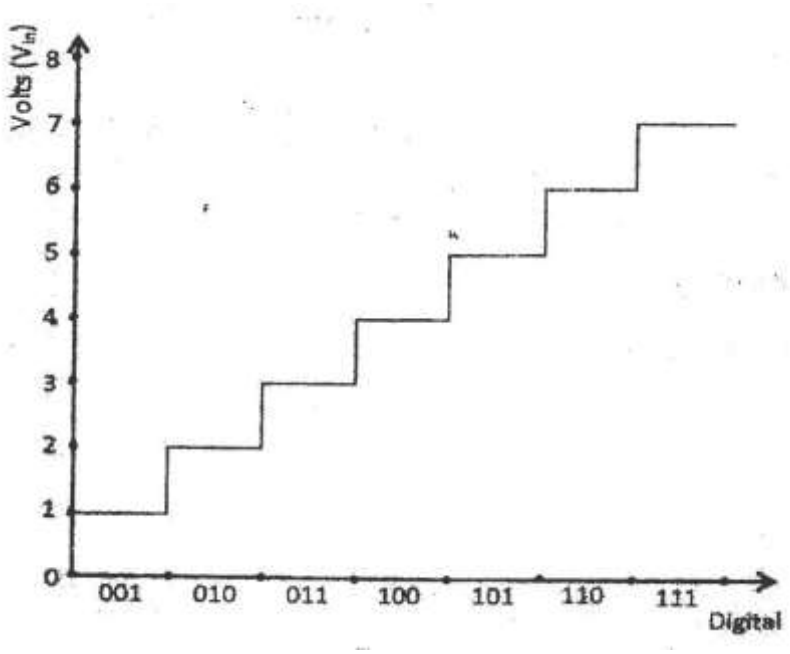


Figure 1: Input output characteristics for a 3-bit quantizer

PROCEDURE

1. A/D Converter

A parallel comparator AID converter is shown in Figure 2.

1. Connect the circuit as shown in the Figure 2, using op-amps, resistors and the encoding unit. V_{in} is the voltage to be converted to the digital form and a, b are the output. 'a' is the most significant bit and 'b' is the least significant bit.
2. Increase the input voltage (V_{in}) from 0V to a voltage around 1V. Observe the output logic level using an oscilloscope. Find the voltage Vin at which output changes from 0,0 to 0,1. Hence fill the following Table 2 (1 represent high logic level and 0 represent low logic level).

Table 2: Voltage V_{in} at which output changes from 0,0 to 0,1

Output a b	**Practical Input Voltage Range(V_{in})**	**Theoretical Inputs Voltage Range(V_{in})**
0 0		
0 1		
1 0		
1 1		

3. Find the resolution of the above system.
4. Explain the behavior of the circuit.
5. Compare the practical and theoretical input voltage ranges and comment.

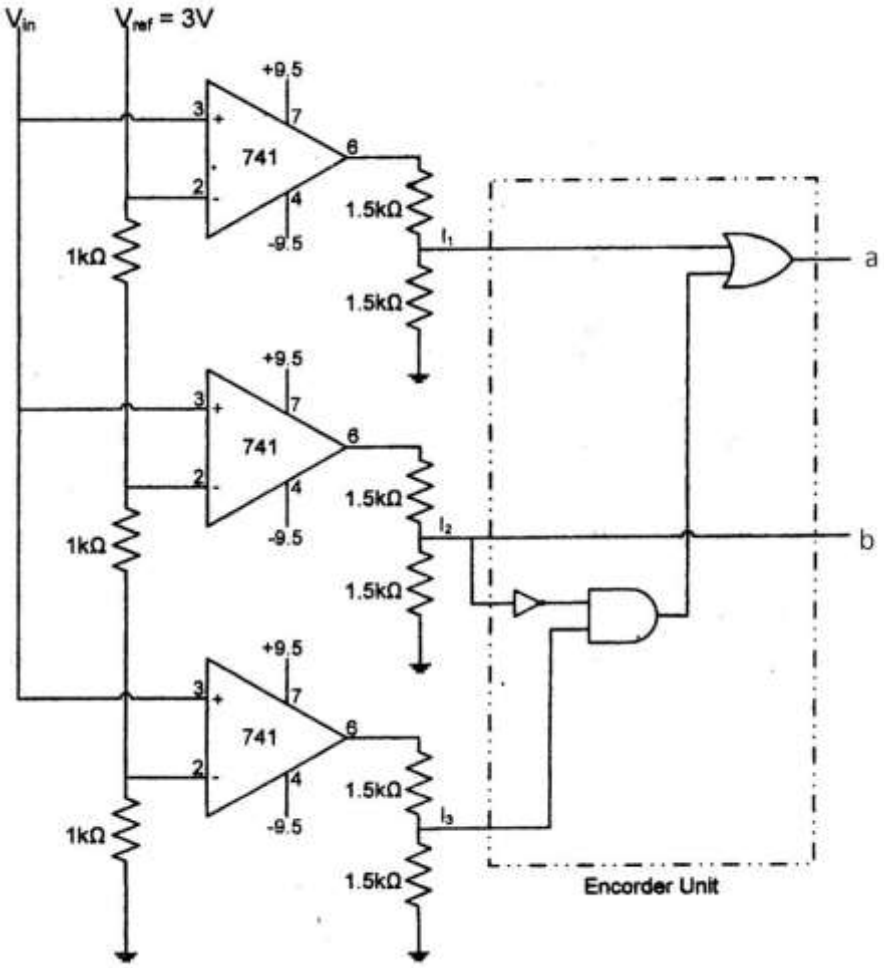


Figure 2: Parallel Comparator A/D converter

2. D/A/Converter

A weighted resistor type D/A converter is shown in Figure 3. Its output is given by the following equation.

$$V_{out} = -V_{ref}\left[\frac{D_0}{R_0} + \frac{D_1}{R_1} + \frac{D_2}{R_2}\right] \times R_f$$

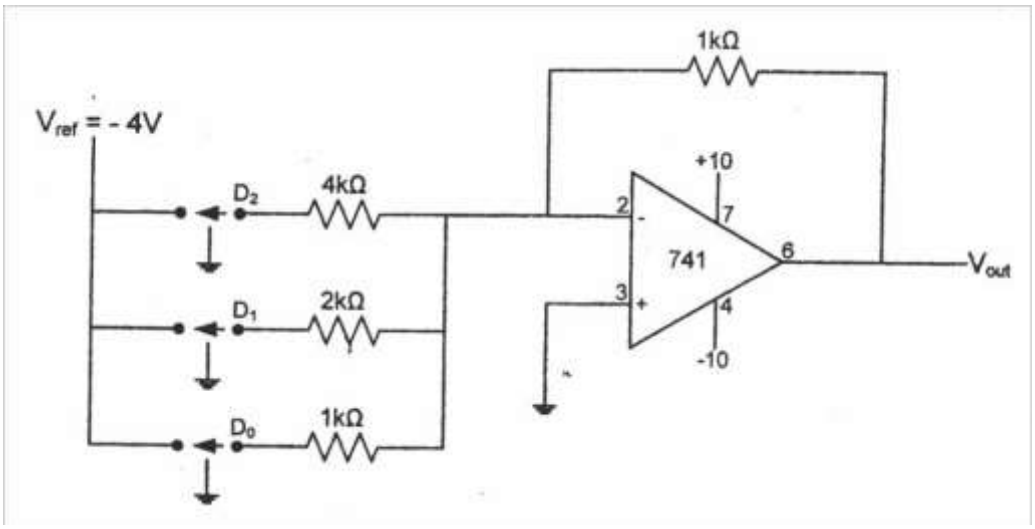


Figure 3: Weighted Resistor D/A Converter D0,

D1, D2 represent 1 and 0

1. Connect the circuit as shown in the Figure 3.
2. Observe V_{out} for each combination of D0, D1 and D2.
3. Prove the above equation from first principals. Calculate theoretical outputs for each combinations of D0, D1 and D2.
4. Compare the theoretical and practical results.

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FACULTY OF ENGINEERING

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EE3305 – Signals and Systems

Laboratory 03 – Analog/Digital Conversion

Laboratory Report

			Signature of the Instructor
Student Number			
Laboratory Conducted Date			
Laboratory Report	**Due Date**		
	Submitted Date		

Evaluation (Instructor Use Only)			
	Evaluation Component	**Allocated Marks Out of 10.0**	**Student Achieved Marks**
1	Pre-Lab Computing (Counter Design)	3.0	
2	Content	4.0	
3	Discussion	3.0	
	Total	10.0	
Signature of the Evaluator			

Pre-Lab Questions :

1. Explain about the analog to digital conversion.

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2. What is the function of PWM in D/A conversion?

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3. What are the Applications of Digital to Analog Converters?

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OBSERVATIONS:

a) A/D Converter

Table 1: Voltage Vin at which output changes from 0,0 to 0,1

Output a b	Practical Input Voltage Range(V_{in})	Theoretical Inputs Voltage Range(V_{in})
0 0		
0 1		
1 0		
1 1		

b) D/A/Converter

Table 2: Observed V_{out} for each combination of D0, D1 and D2

D_0	D_1	D_2	V_{out}

SPECIMAN CALCULATIONS:

b) D/A Conversion

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CALCULATIONS:

a) A/D Conversion

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b) D/A Conversion

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TABULATION:

Table 3: Calculated V_{out} for each combination of D0, D1 and D2

D_0	D_1	D_2	V_{out}

DISCUSSION:

a) A/D Conversion

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b) D/A Conversion

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